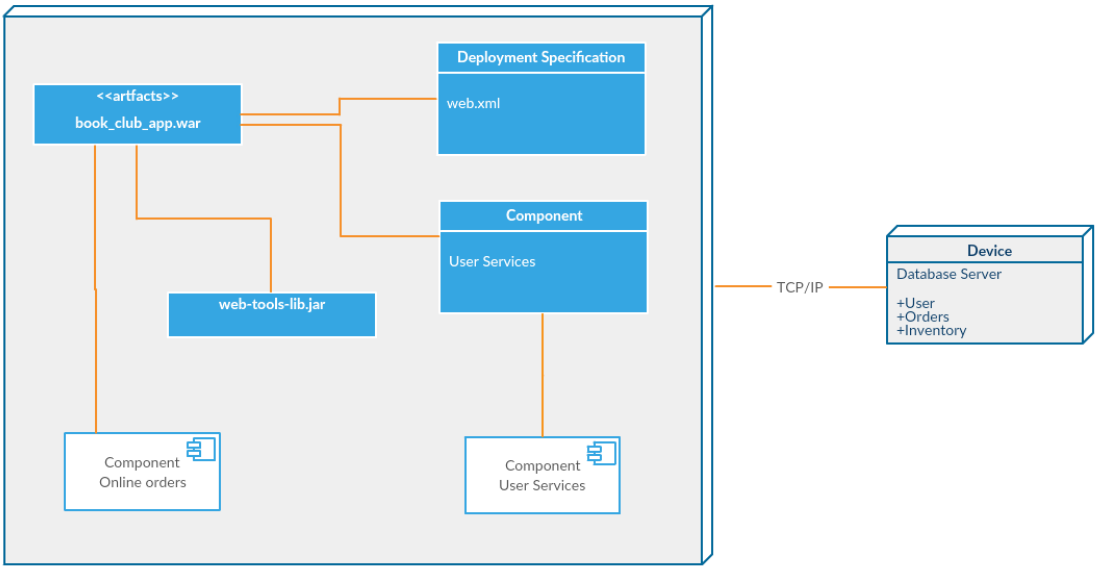
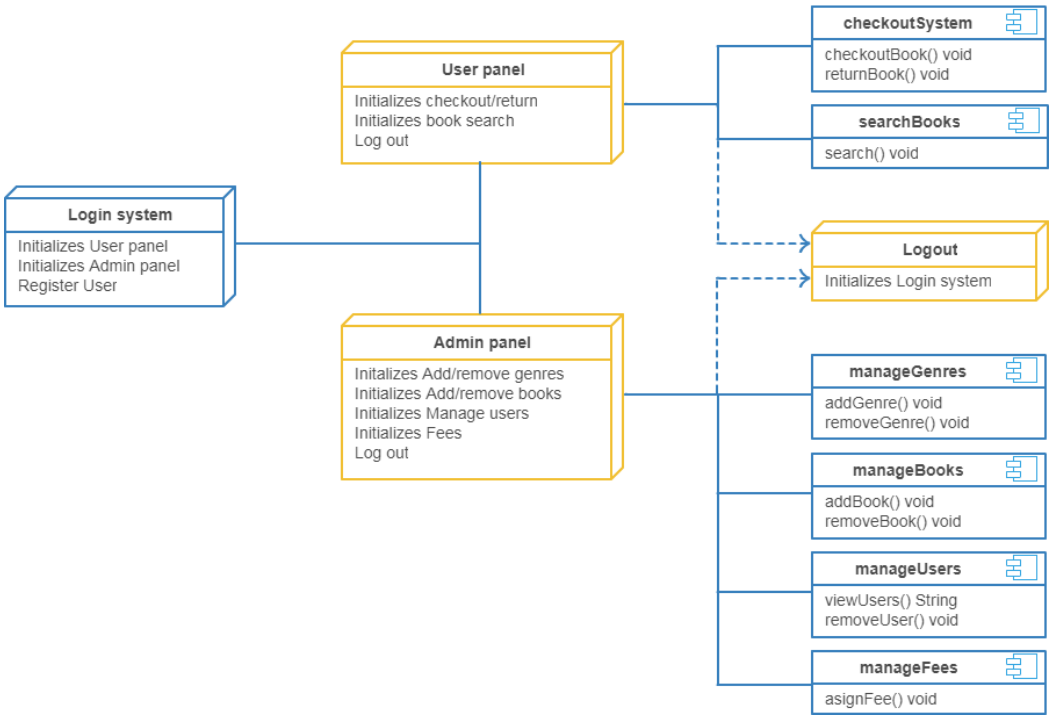
Deployment Diagram for Online Shopping System



Deployment Diagram for Library Management System



Nutan Maharashtra Vidya Prasarak Mandal's

NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of Computer Engineering

LABORATORY MANUAL

SUBJECT: :Laboratory Practice II(AI)

[SUBJECT CODE: CE407]

CLASS: B.E. COMP

PREPARED BY:
Prof. D. S. Shelke

APPROVED BY:

H.O.D. [COMP]

Vision and Mission of the Institute

Vision of the Institute
To be a notable institution for providing quality technical education, ensuring ethical, moral, and holistic development of students.

Mission of the Institute
To nurture engineering graduates with highest technical competence, professionalism and problem solving skills to serve the needs of industry and society.

Vision and Mission of the Department

Department Vision
To foster an environment for technical excellence in Computer Engineering along with holistic growth and ethical responsibility.

Department Mission
M1: To impart engineering knowledge and skills by adopting effective teaching-learning processes.
M2: To develop professional, entrepreneurial and research competencies, fostering continuous intellectual a holistic development.
M3: To nurture students to exhibit societal and ethical responsibility in the professional and social environments.

Program Educational Objectives (PEOs)

1. To provide graduates with a professional career in Electronics, Communication, and allied disciplines.
2. To develop managerial and entrepreneurial skills among the students to solve societal problems.
3. To impart analytical skills in order to develop a multidisciplinary approach to relate engineering issues and innovations.
4. To inculcate effective communication skills, teamwork spirit and professional ethics in students to meet employer needs at large and prepare them for higher studies.

Program Outcomes

PO1: Engineering Knowledge
Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis
Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions
Design creative solutions for complex engineering problems and design/develop systems/components/processes meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems
Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage
Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including predicti and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World
Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its im on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, W and WK7).

PO7: Ethics
Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to nati & international laws. (WK9)

PO8: Individual and Collaborative Team work
Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication
Communicate effectively and inclusively within the engineering community and society at large, such as being a to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance
Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinar environments.

PO11: Life-Long Learning
Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptabi to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK

LP 4 Lab Manual - random

Computer Engineering (Savitribai Phule Pune University)

(digamber.shelke@nmiet.edu.in)

Vision

"Service to society through quality education."

Mission

- Generation of national wealth through education and research
- Imparting quality technical education at the cost affordable to all strata of the society
- Enhancing the quality of life through sustainable development
- Carrying out high-quality intellectual work
- Achieving the distinction of the highest preferred engineering college in theeyes of the stakeholders

Department of Computer Engineering

Vision

"Contributing to the welfare of society through technical and quality education."

Mission

- To produce Best Quality Computer Science Professionals by imparting quality training, hands on experience and value education.
- To Strengthen links with Industry through partnerships and collaborative developmental works.
- To attain self-sustainability and overall development through Research, Consultancy and Development Activities.
- To extend technical expertise to other technical Institutions of the region and play a lead role in imparting technical education."

Programme Outcomes (POs)

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSOs)

Computer Engineering graduate will be able to,

PSO1: **Project Development:** Successfully complete hardware and/or software related system or application projects, using the phases of project development life cycle to meet the requirements of service and product industries; government projects; and automate other engineering stream projects.

PSO2: Domain Expertise: Demonstrate effective application of knowledge gained from different computer domains like, data structures, data bases, operating systems, computer networks, security, parallel programming, in project development, research and higher education.

PSO3: Career Development: Achieve successful Career and Entrepreneurship- The ability to employ modern computer languages, environments, and platforms in creating innovative career paths to be an entrepreneur, and a zest for higher studies.

Laboratory Practice 4 (STQA)

(2019 Pattern)

Practical

Assignments

Part-I: **SOFTWARE TESTING AND QUALITY**

ASSURANCE (STQA)

Part II: Object Oriented Modelling and Design

(OOMD)

Assignment No: 1

Group1:

Aim: Write TEST Scenario for Gmail Login Page

Software Requirements:
Ubuntu 18.04
Windows

Hardware Requirements:
Pentium IV system with latest configuration

Theory:

A **Test Scenario** is defined as any functionality that can be tested. It is also called Test Condition or Test Possibility. As a tester, you should put yourself in the end user's shoes and figure out the real-world scenarios and use cases of the Application Under Test. It is a detailed document of test cases that cover **end to end functionality** of a software application in liner statements. The test scenario is a high-level classification of testable requirements. Before performing the test scenario, the test engineer needs to consider the test cases for each scenario.

In this section, we are going to discuss the difference between the Test Case and Test Scenario as we understood that both the test case and test scenario are part of Testing documentation.

As test engineers, we must be aware of these two testing terms as they play a major role in the Software Test Life Cycle's Test Design and Test Execution phases.

- A test Case is a set of actions executed to verify particular features or functionality, whereas a Test Scenario is any functionality that can be tested.
- Test Case is mostly derived from test scenarios, while Test Scenarios are derived from test artifacts like BRS and SRS.
- Test Case helps in exhaustive testing of an application, whereas Test Scenario helps in an agile way of testing the end-to-end functionality.
- Test Cases are focused on what to test and how to test, while Test Scenario is more focused on what to test.
- Test Cases are low-level actions, whereas Test Scenarios are high-level actions.
- Test Case requires more resources and time for test execution, while Test Scenario requires fewer resources and time for test execution.
- Test Case includes test steps, data, and expected results for testing, whereas Test Scenario includes an end-to-end functionality to be tested.

TEST Scenario for Gmail Login Page:

1) Verify the messages for each mandatory field.
#2) Verify if the user cannot proceed without filling all the mandatory fields.

Sr. No.	Test Scenarios
1	Enter the valid email address & click next. Verify if the user gets an option to enter the password.
2	Don't enter an email address or phone number & just click the Next button. Verify if the user will get the correct message or if the blank field will get highlighted.
3	Enter the invalid email address & click the Next button. Verify if the user will get the correct message.
4	Enter an invalid phone number & click the Next button. Verify if the user will get the correct message.
5	Verify if a user can log in with a valid email address and password.
6	Verify if a user can log in with a valid phone number and password.
7	Verify if a user cannot log in with a valid phone number and an invalid password.
8	Verify if a user cannot log in with a valid email address and a wrong password.
9	Verify the 'Forgot email' functionality.
10	Verify the 'Forgot password' functionality.

#3) Verify the age of the user when the DOB is selected.
#4) Verify if the numbers and special characters are not allowed in the First and Last name.
#5) Verify if a user can sign-up successfully with all the mandatory details.
#6) Verify if a user can log in with the valid details.
#7) Verify if the Password and Confirm Password fields are accepting similar strings only.
#8) Verify if the Password field will prompt you for the weak passwords.
#9) Verify if duplicate email address will not get assigned.
#10) Verify that hints are provided for each field on the form, for the ease of use

Conclusion

Wrote test scenarios on Gmail Login page

Assignment No: 2

Aim: Write Test cases in excel sheet for Social Media application or website

Objectives:
1. Write cases of social media application or website

Software Requirements:
Ubuntu 18.04
Windows

Hardware Requirements:
Pentium IV system with latest configuration

Theory:

The test case is defined as a group of conditions under which a tester determines whether a software application is working as per the customer's requirements or not. Test case designing includes preconditions, case name, input conditions, and expected result. A test case is a first level action and derived from test scenarios.

It is an in-details document that contains all possible inputs (positive as well as negative) and the navigation steps, which are used for the test execution process. Writing of test cases is a one-time attempt that can be used in the future at the time of regression testing.

Test case gives detailed information about testing strategy, testing process, preconditions, and expected output. These are executed during the testing process to check whether the software application is performing the task for that it was developed or not.

Test case helps the tester in defect reporting by linking defect with test case ID. Detailed test case documentation works as a full proof guard for the testing team because if developer missed something, then it can be caught during execution of these full-proof test cases.

To write the test case, we must have the requirements to derive the inputs, and the test scenarios must be written so that we do not miss out on any features for testing. Then we should have the test case template to maintain the uniformity, or every test engineer follows the same approach to prepare the test document.

Generally, we will write the test case whenever the developer is busy in writing the code.

We will write the test case when we get the following:

- When the customer gives the business needs then, the developer starts developing and says that they need 3.5 months to build this product.
- And In the meantime, the testing team will start writing the test cases.
- Once it is done, it will send it to the Test Lead for the review process.

- And when the developers finish developing the product, it is handed over to the testing team.
- The test engineers never look at the requirement while testing the product document because testing is constant and does not depends on the mood of the person rather than the quality of the test engineer.

Test Case ID	**Test Case Description**	**Test Data**	**Test Steps**	**Expected Test Result**	**Actual Test Result**	**Status Pass/Fail**
TC_01	Verify whether a user is able to login with valid username and password	Username : ABCDEFG Password: 123456	1. Open URL : {website homepageurl} 2. Click on Login button from header and navigate to login page 3. Enter registered user's username in the username field 4. Enter valid password 5. Click on Login button	User should be logged in gracefully	Same as Expected	Pass
TC_02	Verify whether it allows user to login with a valid username and an invalid password.	Username : ABCDEFG Password: 1233333	1. Open URL : {website homepageurl} 2. Click on Login button from header and navigate to login page 3. Enter registered user's username in the username field 4. Enter invalid/incorrec	1. It should not allow user tologin 2. It should display validation message for invalid credentials 3. Incorrect input filed should get	Same as expected	Pass

				t password 5. Click on Login button	highlighte d with different color		

					1. It		
	TC_0 3	Verify the login page when the fields are blank and Submit button is clicked.	N/A	1. Open URL : {website homepageurl} 2. Click on Login button from header and navigate to login page 3. Click on Login button	should not allow user tologin 2. Validation message should be display to enter credentials 3. Username and Password field should get highlighte d with different color	Same as expected	Pass

Conclusion

Wrote Test cases in excel sheet for Social Media application or website

Assignment No: 3

Aim: Create Defect Report for Any application or web application

Objectives:

1. Create Defect Report on web application

Software Requirements:
Ubuntu 18.04
Winsows

Hardware Requirements:
Pentium IV system with latest configuration

Theory:

Defect:

A defect in a software product is also known as a bug, error or fault which makes the software produce an unexpected result as per the software requirements. For example; incorrect data, system hangs, unexpected errors, missing or incorrect requirements

Defect Report:

A defect report is a document that has concise details about what defects are identified, what action steps make the defects show up, and what are the expected results instead of the application showing error (defect) while taking particular step by step actions.

Defect reports are usually created by the Quality Assurance team and also by the end-users (customers). Often customers detect more defects and report them to the support team of the software development since the majority of the customers curiously tries out every feature in the application. Now, you know what actually defect and defect reports are.

So let's understand, why do they create defect reports ? What do they do with them?

The reason behind why defect reports are created is to help developers to find out the defects easily and fix them up. A defect report is usually assigned by QA to a developer who then reads the report and reproduces the defects on the software product by following the action steps mentioned in the report. After that, the developer fixes the defects in order to get the desired outcome specified in the report.

That is why the defect reports are important and created carefully. Defect reports should be short, organized, straight to the point and covers all the information that the developer needs to detect the actual defects in the report by doing what and how the one written the defect report detected the defects.

It is usual for QA teams to get defect reports from the clients that are either too short to reproduce and rectify or too long to understand what actually went wrong.

For example,

Defect Description: The application doesn't work as expected.

Now, how in the world does a developer or QA know what went wrong which doesn't meet the client expectation?

In such a case, the developer report to the QA that he couldn't find any problem or he may have fixed any other error but not the actual one client detected. So that's why it's really important to create a concise defect report to get bugs fixed.

All right. You have a pretty good idea about what, whys and how's of a defect report. So it's time for what is inside the report.

Sample bug report 2: Application product Bug report sample

Application testing scenario:

Lets assume in your application you want to create a new user with his/her information, for that you need to logon into the application and navigate to USERS menu > New User, then enter all the details in the User form like, First Name, Last Name, Age, Address, Phone etc. Once you enter all these need to click on SAVE button in order to save the user and you can see a success message saying, "New User has been created successfully".

Now you entered into your application by logging in and navigate to USERS menu > New user, entered all the information and clicked on SAVE button and now the application crashed and you can see one error page on the screen, now you would like to report this BUG.

Now here is how we can report bug for above scenario:

Bug Name: Application crash on clicking the SAVE button while creating a new user.
Bug ID: The BUG Tracking tool will automatically create it once you save this.
Area Path: USERS menu > New Users
Build Number:/Version Number 5.0.1
Severity: HIGH (High/Medium/Low)
Priority: HIGH (High/Medium/Low)

Assigned to: Developer-X
Created By: Your Name
Created On: Date
Reason: Defect

Status: New/Open/Active – Depends on the Tool you are using
Environment: Windows 2003/SQL Server 2005

Description:

Application crash on clicking the SAVE button while creating a new user, hence unable to create a new user in the application.

Steps To Reproduce:

1) Logon into the application
2) Navigate to the Users Menu > New User
3) Filled all the fields
4) Clicked on 'Save' button
5) Seen an error page "ORA1090 Exception: Insert values Error…"
6) See the attached logs for more information

7) And also see the attached screenshot of the error page.

Expected: On clicking SAVE button should be prompted to a success message “New User has been created successfully”.

Save the defect/bug in the BUG TRACKING TOOL.

Conclusion

Hence we learned how to write defect report for the Application

Assignment No: 4

Aim: Installation of Selenium grid and selenium Web driver java eclipse (automation tools).

Objectives:

3. To learn Selenium testing tool
4. To study installation steps of

Software Requirements:
Linux 18.04PHP

Hardware Requirements:
Pentium IV system with latest configuration

Theory:

Selenium is an open source umbrella project for a range of tools and libraries aimed at supporting browser automation. It provides a playback tool for authoring functional tests across most modern web browsers, without the need to learn a test scripting language.

Selenium Grid is a part of the Selenium Suite that specializes in running multiple tests across different browsers, operating systems, and machines in parallel. It is achieved by routing the commands of remote browser instances where a server acts as a hub. A user needs to configure the remote server in order to execute the tests.

When to Use Selenium Grid?
You should use Selenium Grid when you want to do either one or both of following:

Run your tests against different browsers, operating systems, and machines all at the same time. This will ensure that the application you are Testing is fully compatible with a wide range of browser-O.S combinations.
Save time in the execution of your test suites. If you set up Selenium Grid to run, say, 4 tests at a time, then you would be able to finish the whole suite around 4 times faster.

How to Set Up Selenium Grid? Using Command Line

In this section, you will use 2 machines. The first machine will be the system that will run the hub while the other machine will run a node. For simplicity, let us call the machine where the hub runs as “Machine A” while the machine where the node runs will be “Machine B.” It is also important to note their IP addresses. Let us say that Machine A has an IP address of 192.168.1.3 while Machine B has an IP of 192.168.1.4.

Step 1

Download the Selenium Server from here.

Selenium Grid Tutorial: Step by Step Guide with Example
Step 2

You can place the Selenium Server .jar file anywhere in your HardDrive. But for the purpose of this tutorial, place it on the C drive of both Machine A and Machine B. After doing this, you are now done installing Selenium Grid. The following steps will launch the hub and the node.

Step 3

We are now going to launch a hub. Go to Machine A. Using the command prompt, navigate to the root of Machine A's – C drive, because that is the directory where we placed the Selenium Server.
On the command prompt, type java -jar selenium-server-standalone-2.30.0.jar -role hub
The hub should successfully be launched. Your command prompt should look similar to the image below
Selenium Grid Tutorial: Step by Step Guide with Example
Step 4

Another way to verify whether the hub is running is by using a browser. Selenium Grid, by default, uses Machine A's port 4444 for its web interface. Simply open up a browser and go to http://localhost:4444/grid/console

Selenium Grid Tutorial: Step by Step Guide with Example

Also, you can check if Machine B can access the hub's web interface by launching a browser there and going to where "iporhostnameofmachineA" should be the IP address or the hostname of the machine where the hub is running. Since Machine A's IP address is 192.168.1.3, then on the browser on Machine B you should type http://192.168.1.3:4444/grid/console

Step 5

Now that the hub is already set up, we are going to launch a node. Go to Machine B and launch a command prompt there.
Navigate to the root of Drive C and type the code below. We used the IP address 192.168.1.3 because that is where the hub is running. We also used port 5566 though you may choose any free port number you desire.
NOTE: You now have to give path to the Gecko driver if using Firefox. Here is updated code that needs to be used
java -Dwebdriver.gecko.driver="C:\geckodriver.exe" -jar selenium-server-standalone-3.4.0.jar -role webdriver -hub http://192.168.1.3:4444/grid/register -port 5566
Selenium Grid Tutorial: Step by Step Guide with Example

When you press Enter, your command prompt should be similar to the image below.
Selenium Grid Tutorial: Step by Step Guide with Example
Step 6

Go to the Selenium Grid web interface and refresh the page. You should see something like this.

At this point, you have already configured a simple grid. You are now ready to run a test remotely on Machine B.

Designing Test Scripts That Can Run on the Grid

To design test scripts that will run on the grid, we need to use DesiredCapabilites and the RemoteWebDriver objects.

DesiredCapabilites is used to set the type of browser and OS that we will automate
RemoteWebDriver is used to set which node (or machine) that our test will run against.
To use the DesiredCapabilites object, you must first import this package

To use the RemoteWebDriver object, you must import these packages.

Using the DesiredCapabilites Object
Go to the Grid's web interface and hover on an image of the browser that you want to automate. Take note of the platform, and the browserName showed by the tooltip.

In this case, the platform is "XP" and the browserName is "Firefox."

We will use the platform and the browserName in our WebDriver as shown below (of course you need to import the necessary packages first).

Using the RemoteWebDriver Object
Import the necessary packages for RemoteWebDriver and then pass the DesiredCapabilities object that we created above as a parameter for the RemoteWebDriver object.

Running a Sample Test Case on the Grid
Below is a simple WebDriver Testng code that you can create in Eclipse on Machine A. Once you run it, automation will be performed on Machine B.

The test should pass.

 Selenium Grid Tutorial: Step by Step Guide with Example
Selenium grid configuration using JSON File:

JSON stands for Javascript Object Notation. It is a standard format used for interchange of data between browser and the web server. Selenium has an in built JSON config file that can be used to set up selenium grid.

Below are the steps to configure selenium grid using JSON config file.

Step 1) Download the code for JSON config file using the below path

https://github.com/SeleniumHQ/selenium/blob/selenium-3.141.59/java/server/src/org/openqa/grid/common/defaults/DefaultHub.json

Here is the code

```
{
        "port": 4444,
        "newSessionWaitTimeout": -1,
        "servlets" : [],
        "withoutServlets": [],
        "custom": {},
        "capabilityMatcher": "org.openqa.grid.internal.utils.DefaultCapabilityMatcher",
        "registryClass": "org.openqa.grid.internal.DefaultGridRegistry",
        "throwOnCapabilityNotPresent": true,
        "cleanUpCycle": 5000,
        "role": "hub",
        "debug": false,
        "browserTimeout": 0,
        "timeout": 1800
}
```

Step 2) Copy and paste the code on a text editor such as Notepad with the extension '.json'

Step 3)Launch the hub using the below command using command prompt

java -jar selenium-server-standalone-2.53.1.jar -role hub -hubConfig hubconfig.json
NOTE: Selenium stand alone jar file and the json file must be present on the same directory

Step 4) The below screen will appear which indicates the hub is set up successfully

Step 5) Open web browser and try connecting to the below URL
http://192.168.43.223:4444/grid/console
NOTE: The URL may vary from machine to machine. URL followed by 'Nodes should register to' on the above screen must be used.

Step 6) Grid Console screen will appear which indicates the hub is up and running

Configure NODES using JSON:

Step 1) Download the sample node configuration file provided by Selenium using the below URL

https://github.com/SeleniumHQ/selenium/blob/selenium-2.53.0/java/server/src/org/openqa/grid/common/defaults/DefaultNode.json

NOTE: IF YOU ARE USING SELENIUM 3.0 OR ABOVE, THE BELOW JSON FILE MUST BE USED

https://github.com/SeleniumHQ/selenium/blob/selenium-3.141.59/java/server/src/org/openqa/grid/common/defaults/DefaultNodeWebDriver.json

Step 2) Copy and paste the code into a text editor such as Notepad and save the file with the extension '.json'

Step 3)

Run the below command on command prompt

Java-Dwebdriver.chrome.driver="chromedriver.exe" -Dwebdriver.ie.driver="IEDriverServer.exe" -Dwebdriver.gecko.driver="geckodriver.exe" -jar selenium-server-standalone-2.53.1.jar -role node -nodeConfig node1.json
NOTE: Before running the command, please make sure the drivers for each of the browsers have been downloaded onto local machine directory

Step 4) The below screen will appear when enter key is pressed

Step 5) Navigate to the grid console to verify if the nodes have been successfully configured

The browsers are displayed on the grid console. Hence we can be sure that nodes are configured successfully.

Sample Test Cases On Selenium Grid:

```
package com.objectrepository.demo;
import org.openqa.selenium.*;
import org.openqa.selenium.remote.DesiredCapabilities;
import java.net.MalformedURLException;
import java.net.URL;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.*;

public class SeleniumGrid {

    WebDriver driver;
    String baseURL, nodeURL;

    @BeforeTest
    public void setUp() throws MalformedURLException {
       baseURL = "http://demo.guru99.com/test/guru99home/";
```

```
        nodeURL = "http://192.168.43.223:5566/wd/hub";
        DesiredCapabilities capability = DesiredCapabilities.chrome();

        capability.setBrowserName("chrome");
        capability.setPlatform(Platform.WIN10);
        driver = new RemoteWebDriver(new URL(nodeURL), capability);
    }

    @AfterTest
    public void afterTest() {
        driver.quit();
    }
    @Test
    public void sampleTest() {
        driver.get(baseURL);


        if (driver.getPageSource().contains("MOBILE TESTING")) {
            Assert.assertTrue(true, "Mobile Testing Link Found");
        } else {
            Assert.assertTrue(false, "Failed: Link not found");
        }

    }

}
```

The above code launches chrome browser and navigates to the URL specified in the 'baseURL' variable. It verified a link name 'Mobile Testing' is displayed on the page

URL on the 'nodeURL' variable can be modified to reflect the IP Address of the remote machine.

Test result can be verified on the default TestNG report generated

elenium installation is a 3 step process:

Install Java SDK
Install Eclipse
Install Selenium Webdriver Files

In this tutorial, we will learn how to install Selenium Webdriver . Below is the detailed process

NOTE: The versions of Java, Eclipse, Selenium will keep updating with time. But the installation steps will remain the same. Please select the latest version and continue the installation steps below-

Step 1 – Install Java on your computer
Download and install the Java Software Development Kit (JDK) here.

How to Download & Install Selenium
WebDriver Next –

This JDK version comes bundled with Java Runtime Environment (JRE), so you do not need to download and install the JRE separately.

Once installation is complete, open command prompt and type “java”. If you see the following screen you are good to move to the next step

How to Download & Install Selenium
WebDriver Step 2 – Install Eclipse IDE
Download latest version of “Eclipse IDE for Java Developers” here. Be sure to choose correctly between Windows 32 Bit and 64 Bit versions.

You should be able to download an exe file named “eclipse-inst-win64” for Setup.

Double-click on file to Install the Eclipse. A new window will open. Click Eclipse IDE for Java Developers.

After that, a new window will open which click button marked 1 and change path to “C:\eclipse”.
Post that Click on Install button marked 2

After successful completion of the installation procedure, a window will appear. On that window click on Launch

This will start eclipse neon IDE for you.

Step 3 – Download the Selenium Java Client Driver
You can download Selenium Webdriver for Java Client Driver here. You will find client drivers for other languages there, but only choose the one for Java.

This download comes as a ZIP file named “selenium-3.14.0.zip”. For simplicity of Selenium installation on Windows 10, extract the contents of this ZIP file on your C drive so that you would have the directory “C:\selenium-3.14.0\”. This directory contains all the JAR files that we would later import on Eclipse for Selenium setup.

Step 4 – Configure Eclipse IDE with WebDriver
Launch the “eclipse.exe” file inside the “eclipse” folder that we extracted in step 2. If you followed step 2 correctly, the executable should be located on C:\eclipse\eclipse.exe.
When asked to select for a workspace, just accept the default location.

3. Create a new project through File > New > Java Project. Name the project as “newproject”.

A new pop-up window will open enter details as follow
Project Name
Location to save project
Select an execution JRE
Select layout project option
Click on Finish button

4. In this step,

Right-click on the newly created project and

Select New > Package, and name that package as “newpackage”.

A pop-up window will open to name the package,

Enter the name of the package

Click on Finish button

5. Create a new Java class under newpackage by right-clicking on it and then selecting- New > Class, and then name it as “MyClass”. Your Eclipse IDE should look like the image below.

When you click on Class, a pop-up window will open, enter details as
Name of the class
Click on Finish button

This is how it looks like after creating class.

Now selenium WebDriver’s into Java Build Path

In this step,

Right-click on “newproject” and select Properties.
On the Properties dialog, click on “Java Build Path”.
Click on the Libraries tab, and then
Click on “Add External JARs..”

When you click on “Add External JARs..” It will open a pop-up window. Select the JAR files you want to add.

After selecting jar files, click on OK button.

Select all files inside the lib folder.

Select files outside lib folder

How to Download & Install Selenium
WebDriver Once done, click “Apply and Close”
button

How to Download & Install Selenium WebDriver

6. Add all the JAR files inside and outside the “libs” folder. Your Properties dialog should now look similar to the image below.

7. Finally, click OK and we are done importing Selenium libraries into our project.

Different Drivers

HTMLUnit and Firefox are two browsers that WebDriver can directly automate – meaning that no other separate component is needed to install or run while the test is being executed. For other browsers, a separate program is needed. That program is called as the Driver Server.

A driver server is different for each browser. For example, Internet Explorer has its own driver server which you cannot use on other browsers. Below is the list of driver servers and the corresponding browsers that use them.

Conclusion: Hence we learned how to install selenium grid and selenium web driver.

Assignment No: 5

Aim: Prepare Software requirement specification for any project or problem statement

Theory:

Objectives:

1. To Prepare Software requirement specification

Software Requirements:

Windows 7

Hardware Requirements:

Pentium IV system with latest configuration

Introduction

In order to form a good SRS, here you will see some points which can be used and should be considered to form a structure of good SRS. These are as follows :

1. Introduction

- (i) Purpose of this document
- (ii) Scope of this document
- (iii) Overview

2. General description
3. Functional Requirements
4. Interface Requirements
5. Performance Requirements
6. Design Constraints
7. Non-Functional Attributes
8. Preliminary Schedule and Budget
9. Appendices

Software Requirement Specification (SRS) Format as name suggests, is complete specification and description of requirements of software that needs to be fulfilled for successful development of software system. These requirements can be functional as well as non-functional depending upon type of requirement. The interaction between different customers and contractor is done because its necessary to fully understand needs of customers.

Depending upon information gathered after interaction, SRS is developed which describes requirements of software that may include changes and modifications that is needed to be done to increase quality of product and to satisfy customer's demand.

1. **Introduction :**

- **(i) Purpose of this Document –**
  At first, main aim of why this document is necessary and what's purpose of document is explained and described.
- **(ii) Scope of this document –**
  In this, overall working and main objective of document and what value it will provide to customer is described and explained. It also includes a description of development cost and time required.
- **(iii) Overview –**
  In this, description of product is explained. It's simply summary or overall review of product.

2. **General description :**
   In this, general functions of product which includes objective of user, a user characteristic, features, benefits, about why its importance is mentioned. It also describes features of user community.

3. **Functional Requirements :**
   In this, possible outcome of software system which includes effects due to operation of program is fully explained. All functional requirements which may include calculations, data processing, etc. are placed in a ranked order.

4. **Interface Requirements :**
   In this, software interfaces which mean how software program communicates with each other or users either in form of any language, code, or message are fully described and explained. Examples can be shared memory, data streams, etc.

5. **Performance Requirements :**
   In this, how a software system performs desired functions under specific condition is explained. It also explains required time, required memory, maximum error rate, etc.

6. **Design Constraints :**
   In this, constraints which simply means limitation or restriction are specified and explained for design team. Examples may include use of a particular algorithm, hardware and software limitations, etc.

7. **Non-Functional Attributes :**
   In this, non-functional attributes are explained that are required by software system for better performance. An example may include Security, Portability, Reliability, Reusability, Application compatibility, Data integrity, Scalability capacity, etc.

8. **Preliminary Schedule and Budget :**
   In this, initial version and budget of project plan are explained which include overall time duration required and overall cost required for development of project.

9. **Appendices :**
   In this, additional information like references from where information is gathered, definitions of some specific terms, acronyms, abbreviations, etc. are given and explained.

Conclusion: Hence we learnt to design and develop custom Application (Mini Project) using Sales force Cloud

Group 2:

1. Mini Project :Software Testing and Quality Assurance Mini Project Dynamic website of covid19 information using HTML, CSS, JAVASCRIPT And PHP, MySQL database used to store user account, comment, and registration form details. Regular Expression testcases for testing purpose.

2. Mini Project : Mini Project :Create a small application by selecting relevant system environment / platform and programming languages. Narrate concise Test Plan consisting features to be tested and bug taxonomy. Prepare Test Cases inclusive of Test Procedures for identified Test Scenarios. Perform selective Black-box and White-box testing covering Unit and Integration test by using suitable Testing tools. Prepare Test Reports based on Test Pass/Fail Criteria and judge the acceptance of application developed

3. Mini Project : Create a small web-based application by selecting relevant system environment / platform and programming languages. Narrate concise Test Plan consisting features to be tested and bug taxonomy. Narrate scripts in order to perform regression tests. Identify the bugs using Selenium WebDriver and IDE and generate test reports encompassing exploratory testing

Lab Assignment No. 01

Object Oriented Modelling and Design (OOMD), Lab Assignment 1
PROBLEM STATEMENT: Draw state model for telephone line, with various activities.
OBJECTIVES: Demonstrate concept of state chart model for a given problem.

THEORY:

What is a State Machine Diagram?

State Diagram are used to capture the behaviour of a software system. UML State machine diagrams can be used to model the behaviour of a class, a subsystem, a package, or even an entire system. It is also called a state chart or State Transition diagram.

What is a State chart Diagram?

State chart diagrams provide us an efficient way to model the interactions or communication that occur within the external entities and a system. These diagrams are used to model the event-based system. A state of an object is controlled with the help of an event. State chart diagrams are used to describe various states of an entity within the application system.

There are a total of two types of state machine diagram in UML:

1. Behavioural State Machine Diagram

- It captures the behaviour of an entity present in the system.
- It is used to represent the specific implementation of an element.
- The behaviour of a system can be modelled using behavioural state machine diagram in OOAD.

2. Protocol State Machine Diagram

- These diagrams are used to capture the behaviour of a protocol.

- It represents how the state of protocol changes concerning the event. It also represents corresponding changes in the system.
- They do not represent the specific implementation of an element.

Why State Machine Diagram?

State chart diagram is used to capture the dynamic aspect of a system. State machine diagrams are used to represent the behaviour of an application. An object goes through various states during its lifespan. The lifespan of an object remains until the program is terminated. The object goes from multiple states depending upon the event that occurs within the object. Each state represents some unique information about the object.

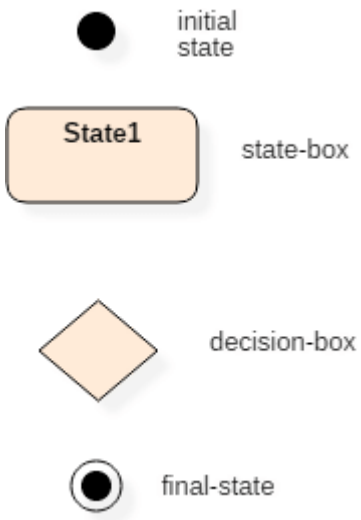
State chart diagrams are used to design interactive systems that respond to either internal or external event. State chart diagram in UML visualizes the flow of execution from one state to another state of an object.

It represents the state of an object from the creation of an object until the object is destroyed or terminated.

The primary purpose of a state chart diagram is to model interactive systems and define each and every state of an object. State chart diagrams are designed to capture the dynamic behaviour of an application system. These diagrams are used to represent various states of a system and entities within the system.

Notation and Symbol for State Machine Diagram (Statechart Diagram)

Following are the various notations that are used throughout the state chart diagram. All these notations, when combined, make up a single diagram.



UML state diagram notations

Initial state

The initial state symbol is used to indicate the beginning of a state machine diagram.

Final state

This symbol is used to indicate the end of a state machine diagram.

Decision box

It contains a condition. Depending upon the result of an evaluated guard condition, a new path is taken for

program execution.

Transition

A transition is a change in one state into another state which is occurred because of some event. A transition causes a change in the state of an object.

State box

It is a specific moment in the lifespan of an object. It is defined using some condition or a statement within the classifier body. It is used to represent any static as well as dynamic situations.

It is denoted using a rectangle with round corners. The name of a state is written inside the rounded rectangle.

The name of a state can also be placed outside the rectangle. This can be done in case of composite or submachine states. One can either place the name of a state within the rectangle or outside the rectangle in a tabular box. One cannot perform both at the same time.

A state can be either active or inactive. When a state is in the working mode, it is active, as soon as it stops executing and transits into another state, the previous state becomes inactive, and the current state becomes active.

Types of State

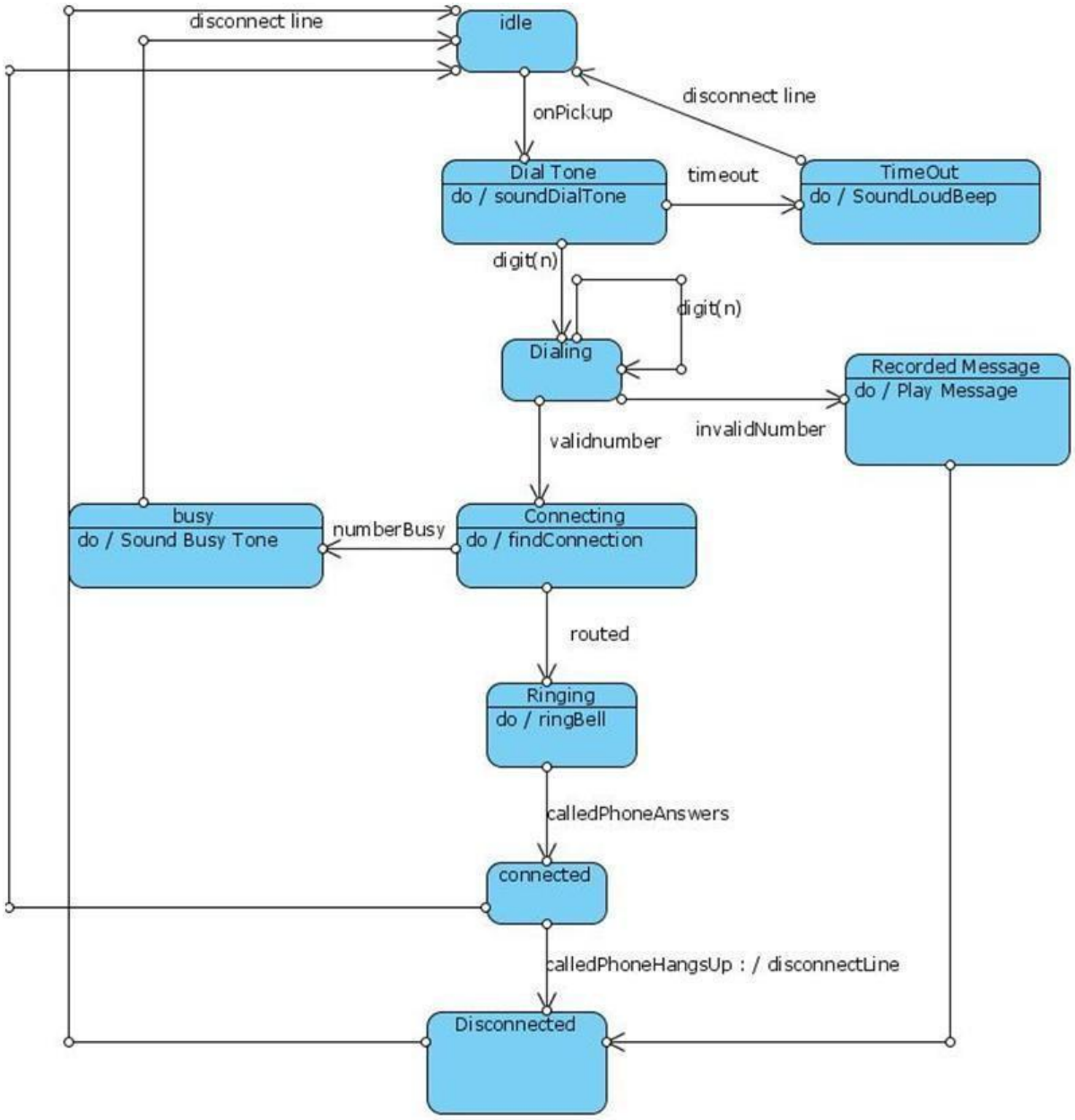
Unified Modelling Language defines three types of states:

- **Simple state**
  - They do not have any substrate.
- **Composite state**
  - These types of states can have one or more than one substrate.
  - A composite state with two or more substates is called an orthogonal state.
- **Submachine state**
  - These states are semantically equal to the composite states.
  - Unlike the composite state, we can reuse the submachine states.

CONCLUSION:

State model for telephone line, with various activitiesCreated successfully.

State model for telephone line



Lab Assignment No. 02

Object Oriented Modelling and Design (OOMD), Lab Assignment 2

PROBLEM STATEMENT:

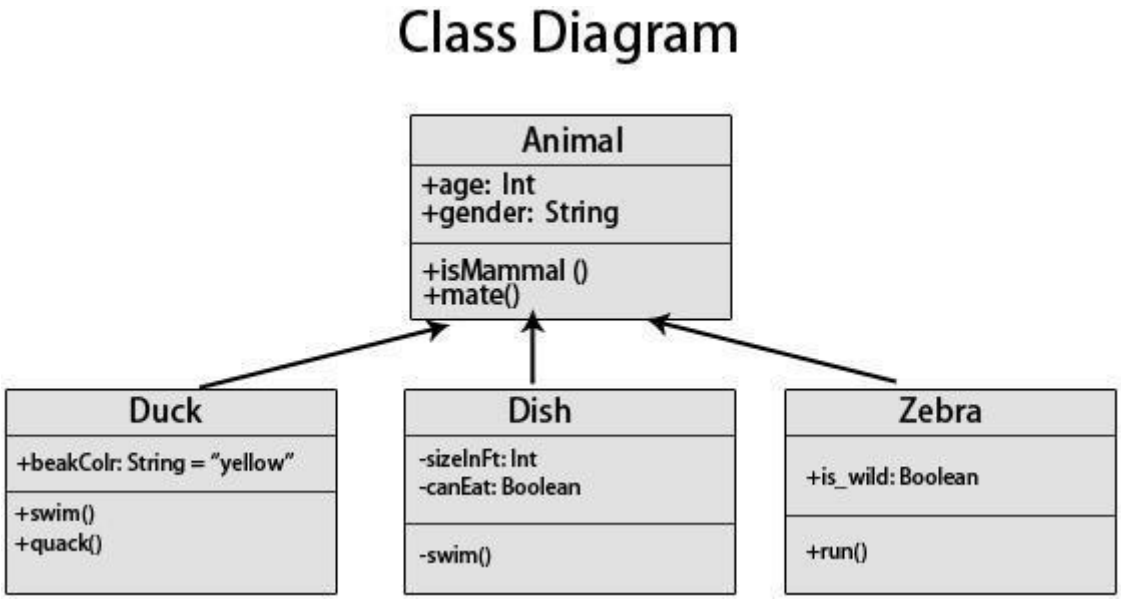
Draw basic class diagrams to identify and describe key concepts like classes, types in your system and their relationships.

OBJECTIVES:

Describe the concepts involved in Object-Oriented modeling class diagrams and their benefits

THEORY:

The class diagram is one of the types of UML diagrams which is used to represent the static diagram by mapping the structure of the systems using classes, attributes, relations, and operations between the various objects. A class diagram has various classes; each has three-part; the first partition contains a Class name which is the name of the class or entity which is participated in the activity, the Second partition contains class attributes that show the various properties of the class, the third partition contains class operations which shows various operations performed by the class, relationships show the relation between two classes.



Relationships

In a class diagram, it is necessary that there exists a relationship between the classes. Unfortunately, the similarity of various relationships often makes it difficult to understand them.

Below are the relationships which exist in a class diagram.

1. Association

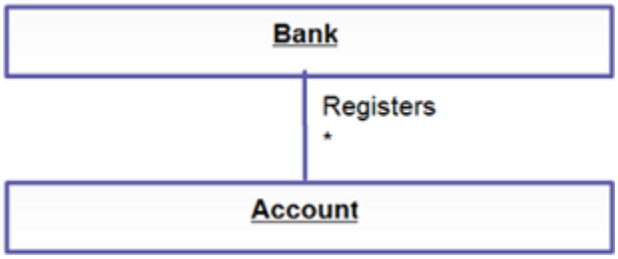
Between two other classes in an association relationship, an association class forms a part of it. Additional information about the relationship could be obtained by attaching the association relationship with the association class. Various operations, attributes, etc., are present in the association class.

The below diagram shows an association between bank and account.



2. Multiplicity

The number of elements or cardinality could be defined by multiplicity. It is one of the most misunderstood relationships which describes the number of instances allowed for a particular element by providing an inclusive non-negative integers interval. It has both lower and upper bound. For example, a bank would have many accounts registered to it. Thus near the account class, a star sign is present.



3. Directed Association

This is a one-directional relationship in a class diagram that ensures the flow of control from one to another classifier. The navigability is specified by one of the association ends. The relationship between two classifiers could be described by naming any association. An arrow indicates the direction of navigation.

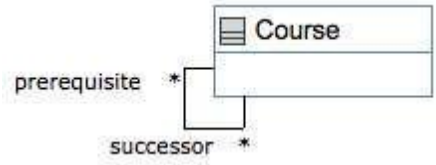
The below example shows an arrowhead relationship between the container and the contained.

This document is available on studocu



4. Reflexive Association

The association of a class to itself is known as Reflexive association, which could be divided into Symmetric and Asymmetric type associations. In Symmetric reflexive association, the semantics of each association end has no logical difference, whereas, in Asymmetric Reflexive Association, the associated class is the same, but there is a semantic difference between the ends of the association.



5. Aggregation

In this type of relationship, a more complex object is created by assembling different objects together. The interaction within the different groups of objects is defined by Aggregation. The integrity of the objects is protected, and the response of the assembled objects is decided by the control object. In aggregation, the classes nurture the 'has a relationship.



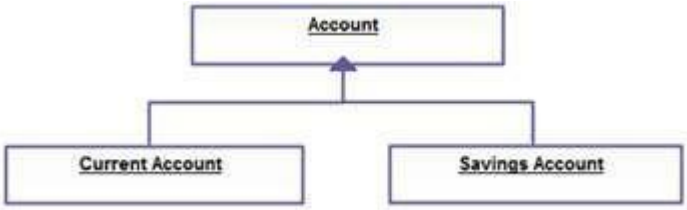
6. Composition

It is a form of aggregation which represents the whole-part relationship. Here, the part classifier lifetime is dependent on the whole classifier lifetime. In a class, a strong life-cycle is represented by the composition relationship. There is usually a one-direction flow of data here. It is generally indicated by a solid line.



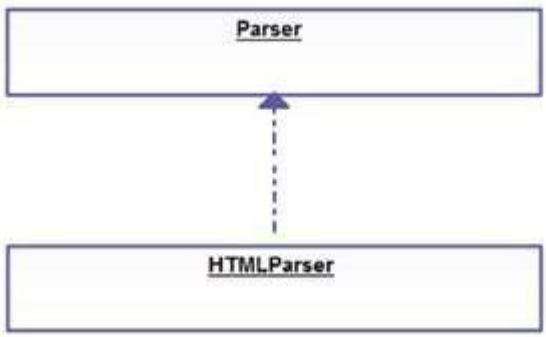
7. Generalization

In this kind of relationship, the child model is based on the parent model. The relationship is used to describe various use-case diagrams and ensures that the child class receives the properties present in the parent. The child model could reuse the attributes of the parent model with the help of the generalization relationship. Hence the distinct attributes need to be defined only in the child; the rest it would inherit from the parent. There could be single parents, multiple children, or multiple parents, single child characteristics in this relationship. There are no names in the generalization relationships. It is also known as the 'is a relationship.



8. Realization

The behaviour of one model element is realized by the specified behaviour of another model element. This type of relationship doesn't have any names.



Why Should we Use Class Diagram?

The structure of a system is defined by a Class Diagram by showing its attributes, relationships among objects, and so on. It is the backbone of object-oriented modelling and could also be used for Data modeling. Class Diagrams help in making pre-plans which eases the programming process. Moreover, you could always make changes to the Class Diagram as it's kind of annoying to code different functionality after facts. It is a design plan based on which a system is built. It is easy to understand without much technical knowledge required.

Class Diagram provides a static view of the application, and its mapping ability with object-oriented language makes it ready to be used in construction. Unlike the sequence diagram, activity diagram, etc., the class diagram is the most popular UML diagram.

Below is the purpose of a Class diagram.

- The static view of an application is designed and analysed.
- A system's responsibilities are described by it.
- The components and the deployment diagram's base is the class diagram.
- The forward and reverse engineering is influenced by the Class Diagram.

Types of Class Diagram

Class Diagram could be divided into three components:

- The Upper Section consists of the class name and is a mandatory component.
- The middle section described the class qualities and used them while describing a class's specific instance.
- The bottom section describes class interaction with the data.

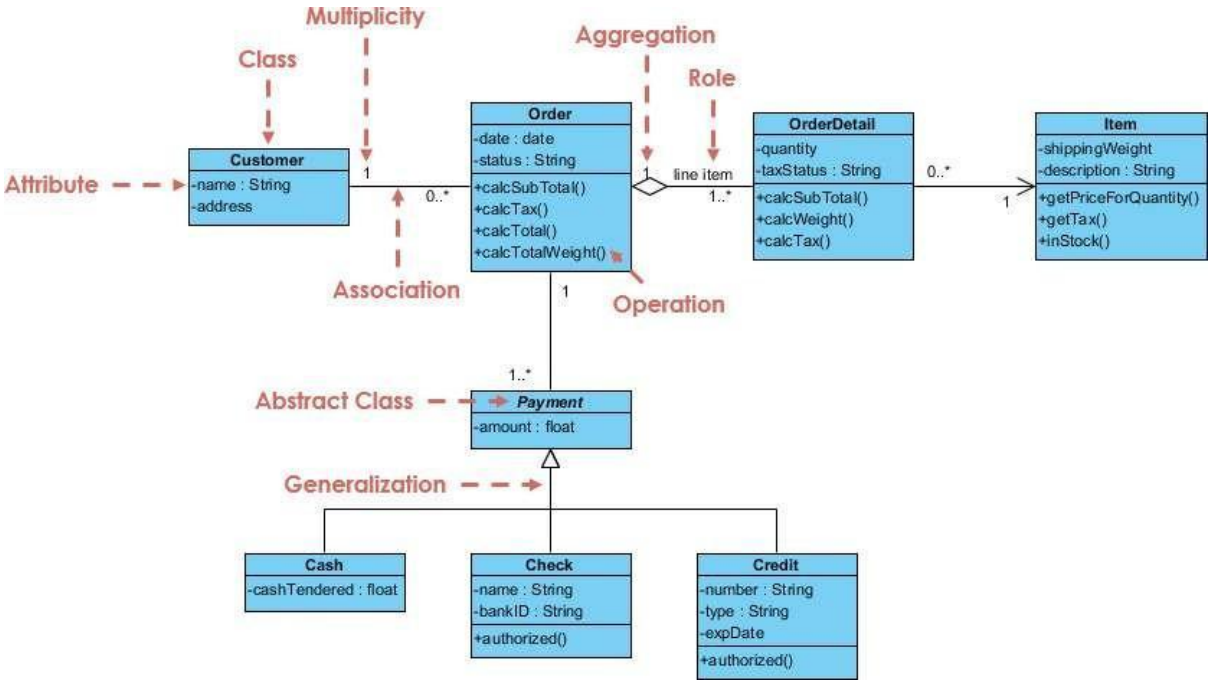
Moreover, a UML is divided into Behavioural and Structural Diagram with Class Diagram falling under the

Structural diagram.

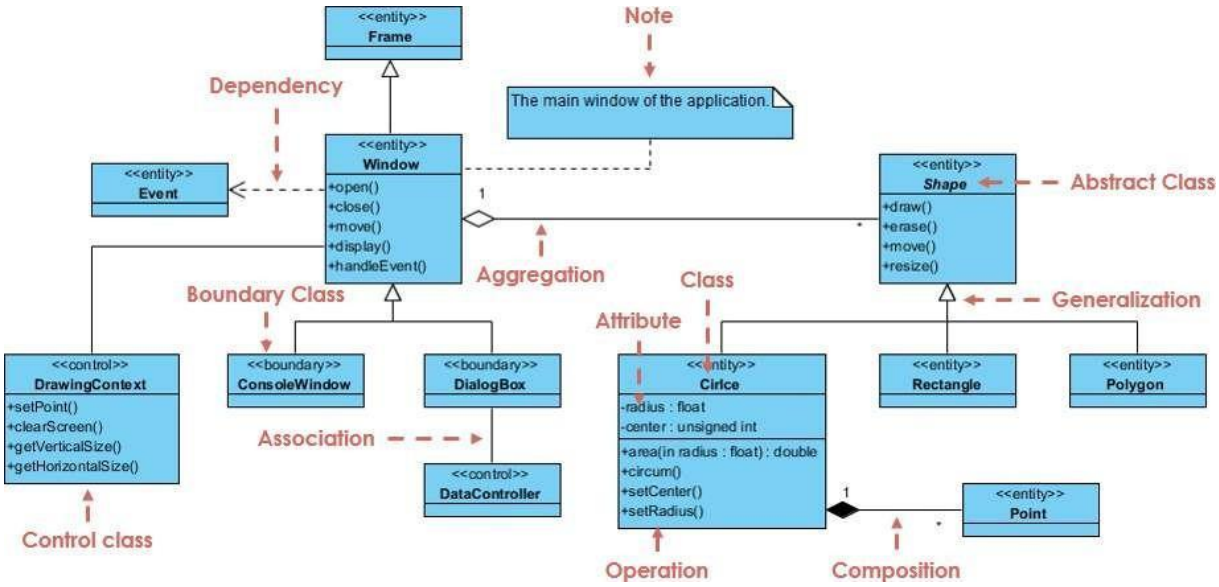
CONCLUSION:

Class diagrams to identify and describe key concepts like classes, types in your system and their relationshipsCreated successfully.

Class Diagram Example: Order System



Class Diagram Example: GUI


<<entity>>
Frame
Note
Dependency
The main window of the application.
<<entity>>
Window
+open()
+close()
+move()
+display()
+handleEvent()
<<entity>>
Event
1
<<entity>>
Shape
+draw()
+erase()
+move()
+resize()
Abstract Class
Aggregation
Boundary Class
Class
Attribute
Generalization
<<control>>
DrawingContext
+setPoint()
+clearScreen()
+getVerticalSize()
+getHorizontalSize()
<<boundary>>
ConsoleWindow
<<boundary>>
DialogBox
<<entity>>
Circle
-radius : float
-center : unsigned int
+area(in radius : float) : double
+circum()
+setCenter()
+setRadius()
<<entity>>
Rectangle
<<entity>>
Polygon
Association
<<control>>
DataController
1
<<entity>>
Point
Control class
Operation
Composition

Lab Assignment No. 03

Object Oriented Modelling and Design (OOMD), Lab Assignment 3

PROBLEM STATEMENT:

Draw one or more Use Case diagrams for capturing and representing requirements of the system. Use case diagrams must include template showing description and steps of the Use Case for various scenarios.

OBJECTIVES:

Demonstrate concept of use-case model, for a given problem.

THEORY:

What is the Use Case Diagram?

Use Case Diagram captures the system's functionality and requirements by using actors and use cases. Use Cases model the services, tasks, function that a system needs to perform. Use cases represent high-level functionalities and how a user will handle the system. Use-cases are the core concepts of Unified Modelling language modelling.

Why Use-Case diagram?

A Use Case consists of use cases, persons, or various things that are invoking the features called as actors and the elements that are responsible for implementing the use cases. Use case diagrams capture the dynamic behaviour of a live system. It models how an external entity interacts with the system to make it work. Use case diagrams are responsible for visualizing the external things that interact with the part of the system.

Use-case:

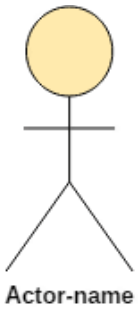
Use cases are used to represent high-level functionalities and how the user will handle the system. A use case represents a distinct functionality of a system, a component, a package, or a class. It is denoted by an oval shape with the name of a use case written inside the oval shape. The notation of a use case in UML is given below:



UML UseCase Notation

Actor:

It is used inside use case diagrams. The actor is an entity that interacts with the system. A user is the best example of an actor. An actor is an entity that initiates the use case from outside the scope of a use case. It can be any element that can trigger an interaction with the use case. One actor can be associated with multiple use cases in the system. The actor notation in UML is given below.



UML Actor Notation

How to draw a use-case diagram?

To draw a use case diagram in UML first one need to analyse the entire system carefully. You have to find out every single function that is provided by the system. After all the functionalities of a system are found out, then these functionalities are converted into various use cases which will be used in the use case diagram.

A use case is nothing but a core functionality of any working system. After organizing the use cases, we have to enlist the various actors or things that are going to interact with the system. These actors are responsible for invoking the functionality of a system. Actors can be a person or a thing. It can also be a private entity of a system. These actors must be relevant to the functionality or a system they are interacting with.

After the actors and use cases are enlisted, then you have to explore the relationship of a particular actor with the use case or a system. One must identify the total number of ways an actor could interact with the system. A single actor can interact with multiple use cases at the same time, or it can interact with numerous use cases simultaneously.

Following rules must be followed while drawing use-case for any system:

1. The name of an actor or a use case must be meaningful and relevant to the system.
2. Interaction of an actor with the use case must be defined clearly and in an understandable way.
3. Annotations must be used wherever they are required.
4. If a use case or an actor has multiple relationships, then only significant interactions must be displayed.

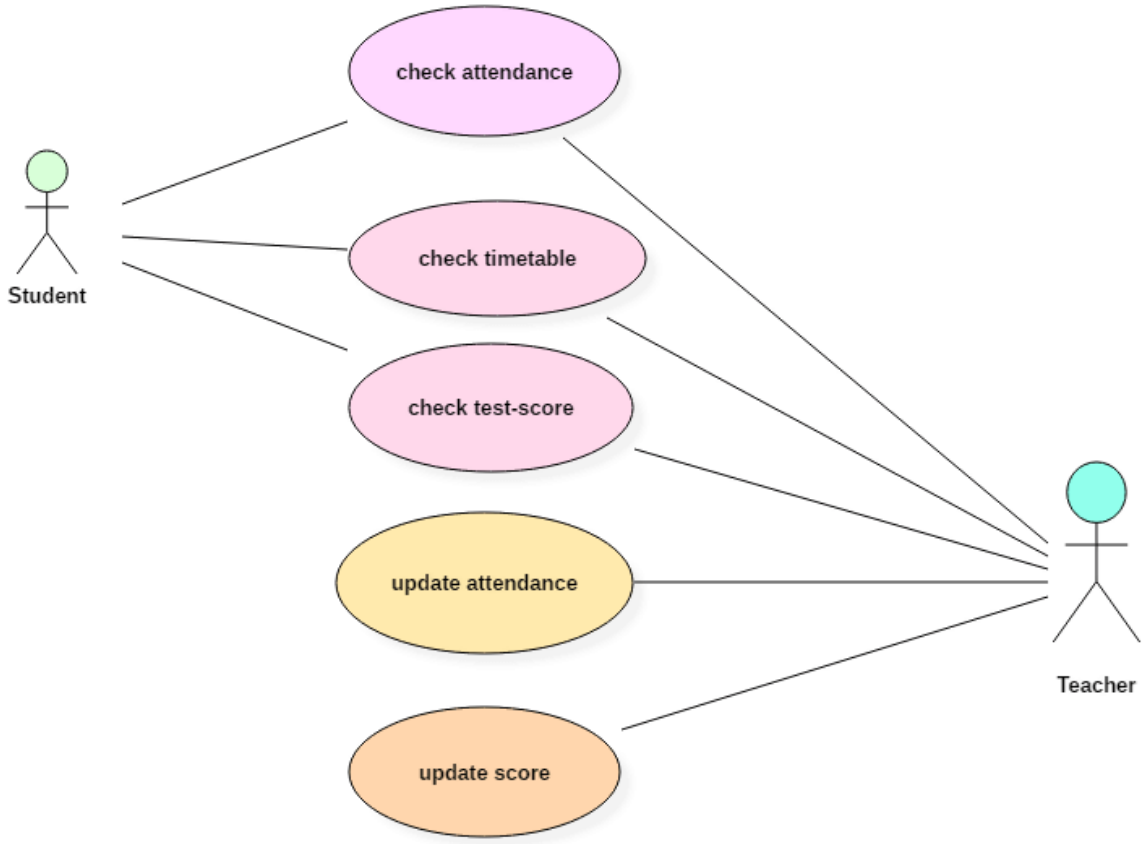
Tips for drawing a use-case diagram

1. A use case diagram should be as simple as possible.
2. A use case diagram should be complete.
3. A use case diagram should represent all interactions with the use case.
4. If there are too many use cases or actors, then only the essential use cases should be represented.
5. A use case diagram should describe at least a single module of a system.
6. If the use case diagram is large, then it should be generalized.

CONCLUSION:

Use Case diagrams for capturing and representing requirements of the systemcreated successfully.

Use Case diagrams



Lab Assignment No. 05

Object Oriented Modelling and Design (OOMD), Lab Assignment 5
PROBLEM STATEMENT: Draw activity diagrams to display either business flows or like flow charts
OBJECTIVES: Demonstrate concept activity diagrams are a variation for a given problem.

THEORY:

Activity diagrams can be used in all stages of software development and for various purposes. And because they are a lot similar to flowcharts, they are generally more popular than other UML diagram types.

What is an activity diagram?

A UML activity diagram helps to visualize a certain use case at a more detailed level. It is a behavioural diagram that illustrates the flow of activities through a system.

UML activity diagrams can also be used to depict a flow of events in a business process. They can be used to examine business processes in order to identify its flow and requirements.

How to Draw an Activity Diagram?

Activity diagrams are mainly used as a flowchart that consists of activities performed by the system. Activity diagrams are not exactly flowcharts as they have some additional capabilities. These additional capabilities include branching, parallel flow, swimlane, etc.

Before drawing an activity diagram, we must have a clear understanding about the elements used in activity diagram. The main element of an activity diagram is the activity itself. An activity is a function performed by the system. After identifying the activities, we need to understand how they are associated with constraints and conditions.

Before drawing an activity diagram, we should identify the following elements −

- Activities
- Association

- Conditions
- Constraints

Once the above-mentioned parameters are identified, we need to make a mental layout of the entire flow. This mental layout is then transformed into an activity diagram.

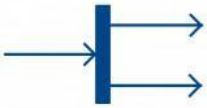
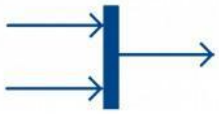
Following is an example of an activity diagram for order management system. In the diagram, four activities are identified which are associated with conditions. One important point should be clearly understood that an activity diagram cannot be exactly matched with the code. The activity diagram is made to understand the flow of activities and is mainly used by the business users

`Following diagram is drawn with the four main activities –`

- Send order by the customer
- Receipt of the order
- Confirm the order
- Dispatch the order

Activity Diagram Symbols

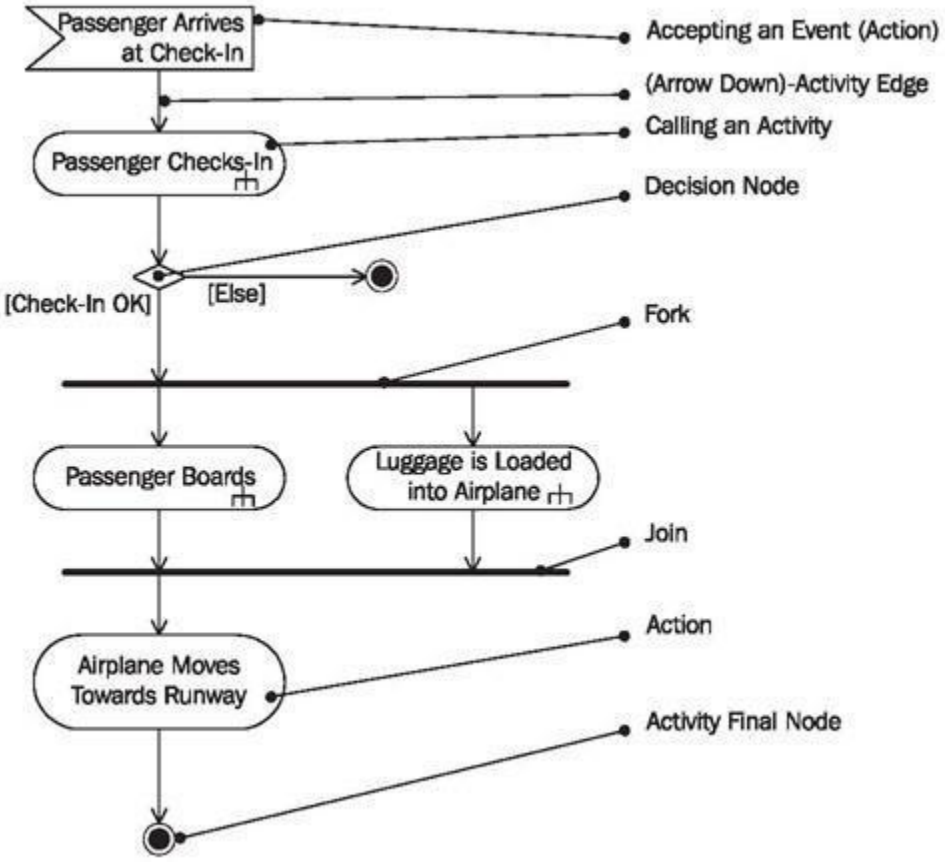
Symbol	Name	Use
	Start/ Initial Node	Used to represent the starting point or the initial state of an activity
	Activity / Action State	Used to represent the activities of the process
	Action	Used to represent the executable sub-areas of an activity
	Control Flow / Edge	Used to represent the flow of control from one action to the other
	Object Flow / Control Edge	Used to represent the path of objects moving through the activity
	Activity Final Node	Used to mark the end of all control flows within the activity
	Flow Final Node	Used to mark the end of a single control flow

	Decision Node	Used to represent a conditional branch point with one input and multiple outputs
	Merge Node	Used to represent the merging of flows. It has several inputs, but one output.
	Fork	Used to represent a flow that may branch into two or more parallel flows
	Merge	Used to represent two inputs that merge into one output
	Signal Sending	Used to represent the action of sending a signal to an accepting activity
	Signal Receipt	Used to represent that the signal is received

CONCLUSION:

Activity diagrams to display either business flows or like flow charts created successfully.

Activity diagram


Passenger Arrives at Check-In
Accepting an Event (Action)
(Arrow Down)-Activity Edge
Calling an Activity
Passenger Checks-In
Decision Node
[Check-In OK]
[Else]
Fork
Passenger Boards
Luggage is Loaded into Airplane
Join
Action
Airplane Moves Towards Runway
Activity Final Node

Lab Assignment No. 06

Object Oriented Modelling and Design (OOMD), Lab Assignment 6

PROBLEM STATEMENT:

Draw component diagrams assuming that you will build your system reusing existing components along with a few new ones.

OBJECTIVES:

Demonstrate concept of component diagrams for a given problem.

THEORY:

While other UML diagrams, which describe the functionality of a system, component diagrams are used to model the components that help make those functionalities.

In this component diagram tutorial, we will look at what a component diagram is, component diagram symbols, and how to draw one. You can use a component diagram example below to get a quick start.

What is Component Diagram

Component diagrams are used to visualize the organization of system components and the dependency relationships between them. They provide a high-level view of the components within a system.

The components can be a software component such as a database or user interface; or a hardware component such as a circuit, microchip or device; or a business unit such as supplier, payroll or shipping.

Component diagrams

- Are used in Component-Based-Development to describe systems with Service-Oriented-Architecture
- Show the structure of the code itself
- Can be used to focus on the relationship between components while hiding specification detail
- Help communicate and explain the functions of the system being built to stakeholders

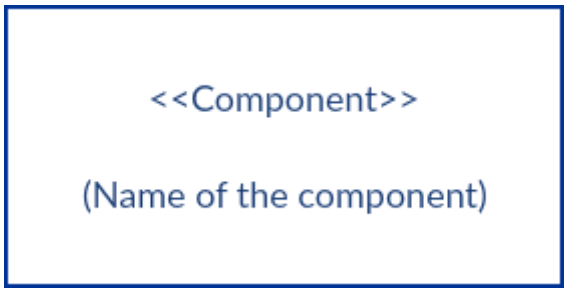
Component Diagram Symbols

We have explained below the common component diagram notations that are used to draw a component diagram.

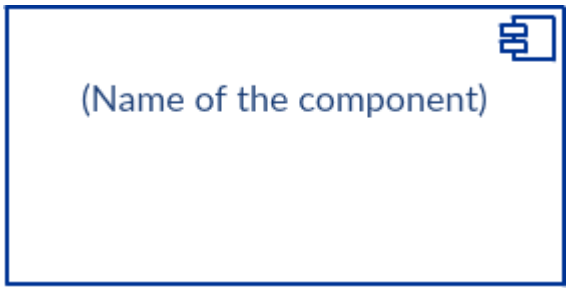
Component

There are three ways the component symbol can be used.

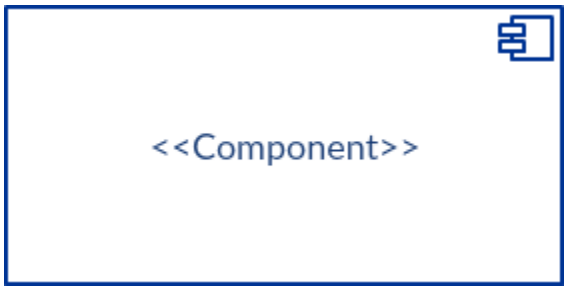
1) Rectangle with the component stereotype (the text <<component>>). The component stereotype is usually used above the component name to avoid confusing the shape with a class icon.



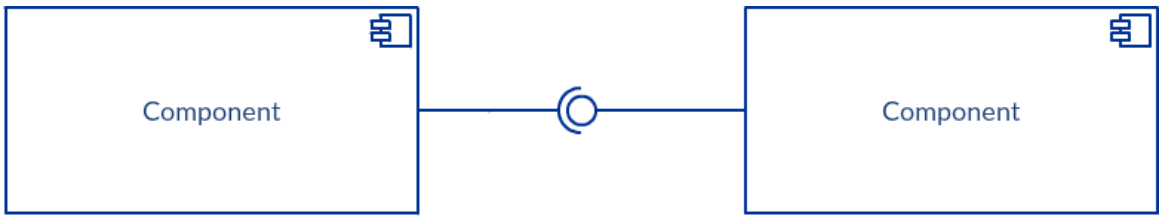
2) Rectangle with the component icon in the top right corner and the name of the component.



3) Rectangle with the component icon and the component stereotype.

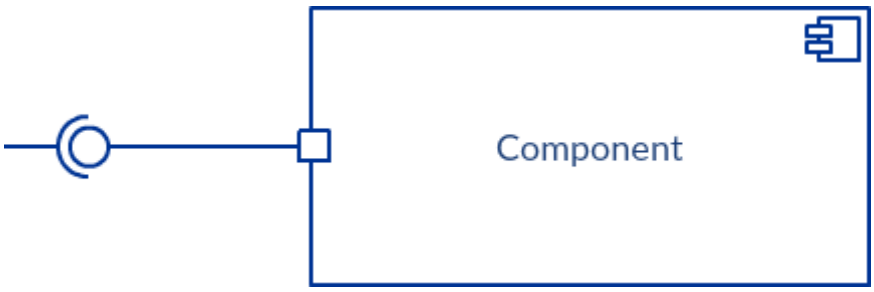


Provided Interface and the Required Interface



Interfaces in component diagrams show how components are wired together and interact with each other. The assembly connector allows linking the component's required interface (represented with a semi-circle and a solid line) with the provided interface (represented with a circle and solid line) of another component. This shows that one component is providing the service that the other is requiring.

Port



Port (represented by the small square at the end of a required interface or provided interface) is used when the component delegates the interfaces to an internal class.

Dependencies



Although you can show more detail about the relationship between two components using the ball-and-socket notation (provided interface and required interface), you can just as well use a dependency arrow to show the relationship between two components.

How to Draw a Component Diagram

You can use a component diagram when you want to represent your system as components and want to show their interrelationships through interfaces. It helps you get an idea of the implementation of the system. Following are the steps you can follow when drawing a component diagram.

Step 1: figure out the purpose of the diagram and identify the artifacts such as the files, documents etc. in your system or application that you need to represent in your diagram.

Step 2: As you figure out the relationships between the elements you identified earlier, create a mental layout of your component diagram

Step 3: As you draw the diagram, add components first, grouping them within other components as you see fit

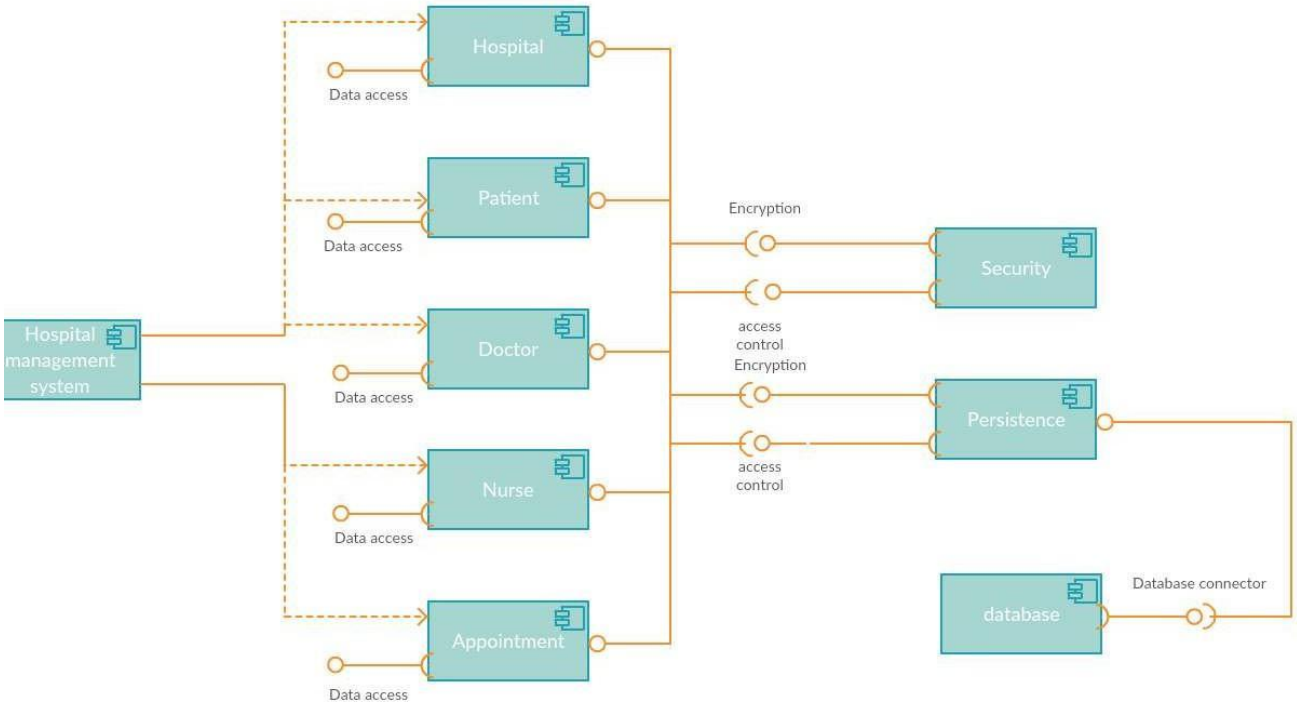
Step 4: Next step is to add other elements such as interfaces, classes, objects, dependencies etc. to your component diagram and complete it.

Step 5: You can attach notes on different parts of your component diagram to clarify certain details to others.

CONCLUSION:

State model for telephone line, with various activities Created successfully.

COMPONENT DIAGRAM OF HOSPITAL MANAGEMENT SYSTEM



Lab Assignment No. 07

Object Oriented Modelling and Design (OOMD), Lab Assignment 7
PROBLEM STATEMENT: Draw deployment diagrams to model the runtime architecture of your system
OBJECTIVES: Demonstrate concept of deployment diagrams for a given problem.

THEORY:

Deployment diagram in the Unified Modeling Language models the *physical* deployment of artifacts on nodes.[1] To describe a web site, for example, a deployment diagram would show what hardware components ("nodes") exist (e.g., a web server, an application server, and a database server), what software components ("artifacts") run on each node (e.g., web application, database), and how the different pieces are connected (e.g. JDBC, REST, RMI).

The nodes appear as boxes, and the artifacts allocated to each node appear as rectangles within the boxes. Nodes may have subnodes, which appear as nested boxes. A single node in a deployment diagram may conceptually represent multiple physical nodes, such as a cluster of database servers.

There are two types of Nodes:

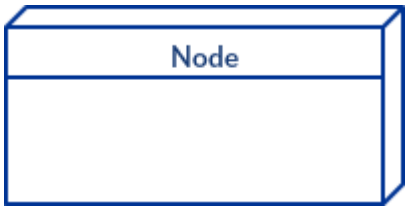
1. Device Node
2. Execution Environment Node

Device nodes are physical computing resources with processing memory and services to execute software, such as typical computers or mobile phones. An execution environment node (EEN) is a software computing resource that runs within an outer node and which itself provides a service to host and execute other executable software elements.

Deployment Diagram Notations

In order to draw a deployment diagram, you need to first become familiar with the following deployment diagram notations and deployment diagram elements.

Nodes



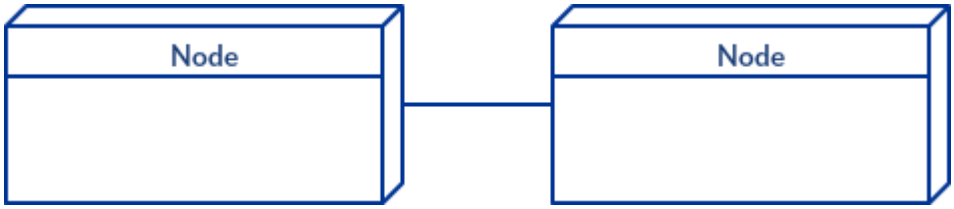
A node, represented as a cube, is a physical entity that executes one or more components, subsystems or executables. A node could be a hardware or software element.

Artifacts



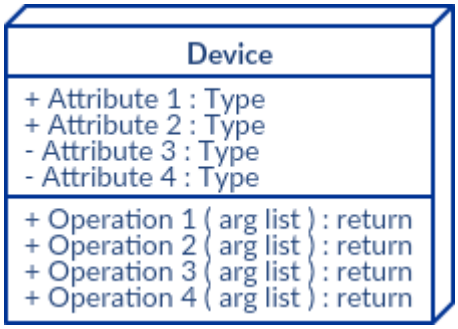
Artifacts are concrete elements that are caused by a development process. Examples of artifacts are libraries, archives, configuration files, executable files etc.

Communication Association



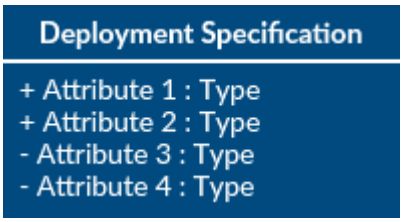
This is represented by a solid line between two nodes. It shows the path of communication between nodes.

Devices



A device is a node that is used to represent a physical computational resource in a system. An example of a device is an application server.

Deployment Specifications



Deployment specifications is a configuration file, such as a text file or an XML document. It describes how an artifact is deployed on a node.

How to Draw a Deployment Diagram

Follow the simple steps below to draw a deployment diagram. You can either use the deployment diagram examples below to get a head start or use our UML diagram tool to start from the beginning.

Step 1: Identify the purpose of your deployment diagram. And to do so, you need to identify the nodes and devices within the system you'll be visualizing with the diagram.

Step 2: Figure out the relationships between the nodes and devices. Once you know how they are connected, proceed to add the communication associations to the diagram.

Step 3: Identify what other elements like components, active objects you need to add to complete the diagram.

Step 4: Add dependencies between components and objects as required.